

components will be affected by changing the CSS values.

- **Cross-platform:** The most interesting thing about Scene Builder is that it is written as a self-contained JavaFX application and supports the Windows, Mac and Linux operating systems.

Beginners can benefit a lot from Scene Builder because it reduces the effort required to build user interfaces considerably. So time and energy can be more productively focused on the application logic, and on its look and feel. Table 1 lists some of the differences between the visual approach and the coded approach for building a UI.

Table 1

Visual approach	Coded approach
Fast approach towards development.	Development is slow.
Unused functions can be left due to the auto generation facility.	Only the required and appropriate functions will be available.
One can instantly get an idea of the look and feel of the interface; so modifications are easy.	Modifying the UI is time consuming and difficult, requiring many man hours.

Sections of Scene Builder

There are different sections within Scene Builder, each with its own functionality. Broadly, it can be divided into four sections, as listed below.

1. **Library section:** This is the section from where we can drag and drop different widgets to the interface. It contains a search text box, where one can search for different controls, shapes, etc. All the widgets are classified into the following seven areas:
 - **Containers** – This contains different panes such as AnchorPane, VBox, HBox, BorderPane, etc. These are mainly used to store the different controls.
 - **Controls** – This contains the main components used to build your application such as buttons, checkboxes, drop down menus, date pickers, etc.
 - **Menu** – This area contains different menu items such as CheckMenuItem, ContextMenu, MenuItem, etc.
 - **Miscellaneous** – Canvas, Tool tip, etc, are the widgets present in this area.
 - **Shapes** – This area is used to draw different shapes, like a circle, arc, ellipse, etc.
 - **Charts** – Various types of charts can be dragged from this area such as AreaChart, BarChart, etc.
 - **3D** – This area is mainly used if you are working with 3D objects in your application.
2. **Document section:** This contains two areas—hierarchy and controller. The former contains the hierarchy of your components in your interface, and the latter is used to connect your Scene Builder to the controller class in your

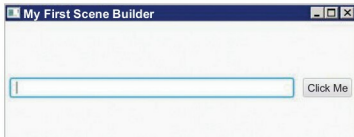


Figure 1: This is what the first application you've designed will look like

project. We will explore controller class and its usage later.

3. **Centre section:** This is the working section, where you drop the dragged components and view your application interface.
4. **Inspector section:** This section contains three areas—properties, layout and code. In the properties area, we can define the different properties of our components; the layout area deals with the positioning of components in the interface, and the code area contains the logic behind each component.

Downloading and configuring Scene Builder

The Scene Builder project was started by Oracle so that organisations could move to JavaFX quickly and installers could be downloaded from Oracle's website. But after the 40th update of Java 8, Oracle stopped development and declared that Scene Builder would only be available as source code and no installers would be provided. Then an organisation named Gluon came to the rescue of the developer community and started providing installers for Scene Builder. Though Oracle's version is still available under archive projects, it's Gluon's Scene Builder that is popular due to its regular updates, and can be downloaded from Gluon's website <http://gluonhq.com/> depending on the platform required. Also, those already using Oracle's Scene Builder needn't worry, since it will be automatically updated.

 **Note:** Gluon started providing support for Scene Builder after Java 8 update 40; so you need to have JDK 8 installed on your system in order to use it.

Now let's take a look at how Scene Builder can be configured with Eclipse. First, you should have Scene Builder and the Eclipse IDE downloaded on your system.

 **Note:** All the images and screenshots in this article are based on the Eclipse Neon 64-bit version.

Configuring Scene Builder

1. Scene Builder will be downloaded as an .exe file, and you have to install it by double-clicking it. Follow the steps 2 to 10 given below and Scene Builder will be installed within minutes. After the installation is complete, an instance will be opened. Though Scene Builder has been installed, it has not yet been configured